



Python for Fixed Income

COURSE 7 · NLP AND LLMS

Case Study: FOMC Statement Tone over Time

WEEK 1 · DAY 5

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Two numbers per statement

Hands-on: read a statement

The course's two word lists and `dictionary_score`, a count per thousand tokens. Then `policy_action`, the action a statement states, checked against the rate it states.

Case study: 213 scheduled statements

A dictionary score and a classifier's score for each statement, the classifier fitted only on earlier years, with the label's own words taken out first.

It closes with: what the model shows, and where it fails.

Real documents, described

A statement's count or score

Allowed names

count of words from the course's list

dictionary score

the action the statement states

Never called

hawkish or dovish

a signal or a policy view

a forecast of rates

The classifier's output today

Allowed names

the classifier's score for raise

the label the model assigns

a walk-forward macro F1

Never called

a probability, with class weights

what the Committee meant

a reading of policy

The course's word lists

Firming

raise, increase, tighten, firming, elevated

Easing

lower, reduce, accommodation, accommodative, easing

```
score = (firming - easing) / tokens * 1000    # points per thousand tokens  
"... did not lower it." -> lower still counts: 58.8235
```

Named and not used: VADER (its authors: attuned to social media) and Loughran-McDonald (its page: free for use in academic research).

The lists are a choice, one of many: printed in full.

The action each statement states

139

hold

47

raise, by policy_action

27

lower

219

checked against the stated rate

213

agree

6

disagree, all in 2015

In each of the six, the first sentence the pattern reads is about a raise at a later date; the range stays at 0 to 1/4 percent.

Giving a label is not giving the right one. Case study label: 41 raise, 172 other.

The case study

How did a count of the course's words move from 2000 to 2026, and can a classifier fitted only on earlier statements tell, from the rest of the text, which announced a raise?

1. Take the label's own words out first, after showing the leak
2. A dictionary score for every statement
3. The classifier, walk-forward by year from 2005: 173 scored
4. Side by side, with a bootstrap interval, and a workbook
5. What the model shows, and where it fails

40 statements of 2000 to 2004 are fitted on only. 173 statements in 22 years are scored, 33 of them raises.

The label is in the text

Term (action left in)	Coefficient
raise	0.8956
to raise	0.8956
rate to	0.6562
raise the	0.6501
raise its	0.4923
action the	0.4920

Walk-forward macro F1, leak in: 0.6562

The label's own words top the coefficients, yet the score is not near 1.0. A leak does not need to lift the score to be a leak.

467

sentences dropped

0

kept texts that still state an action

0.8046

share of tokens kept

Two numbers per statement

Word	List	Count
raise	firming	0
increase	firming	65
tighten	firming	0
firming	firming	25
elevated	firming	104
lower	easing	38
reduce	easing	16
accommodation	easing	90
accommodative	easing	156
easing	easing	15

Model	Macro F1	Accuracy	Recall
majority_label	0.4473	0.8092	0.0000
tfidf_logistic_balanced_action_left_in	0.6562	0.7688	0.5152
tfidf_logistic_balanced	0.6510	0.7630	0.5152

Walk-forward on the kept text: 0.6510 against 0.4473 for the majority label.

Raise and tighten are gone with the rate sentences; elevated and accommodative carry the score.

A second, smaller leak

Term (kept text)	Coefficient
action the	0.5036
cleveland richmond	0.4404
related	0.4338
approved	0.4172

In a related action...

...the Board of Governors approved an increase in the discount rate: beside a change in the target, and it names no federal funds rate.

20

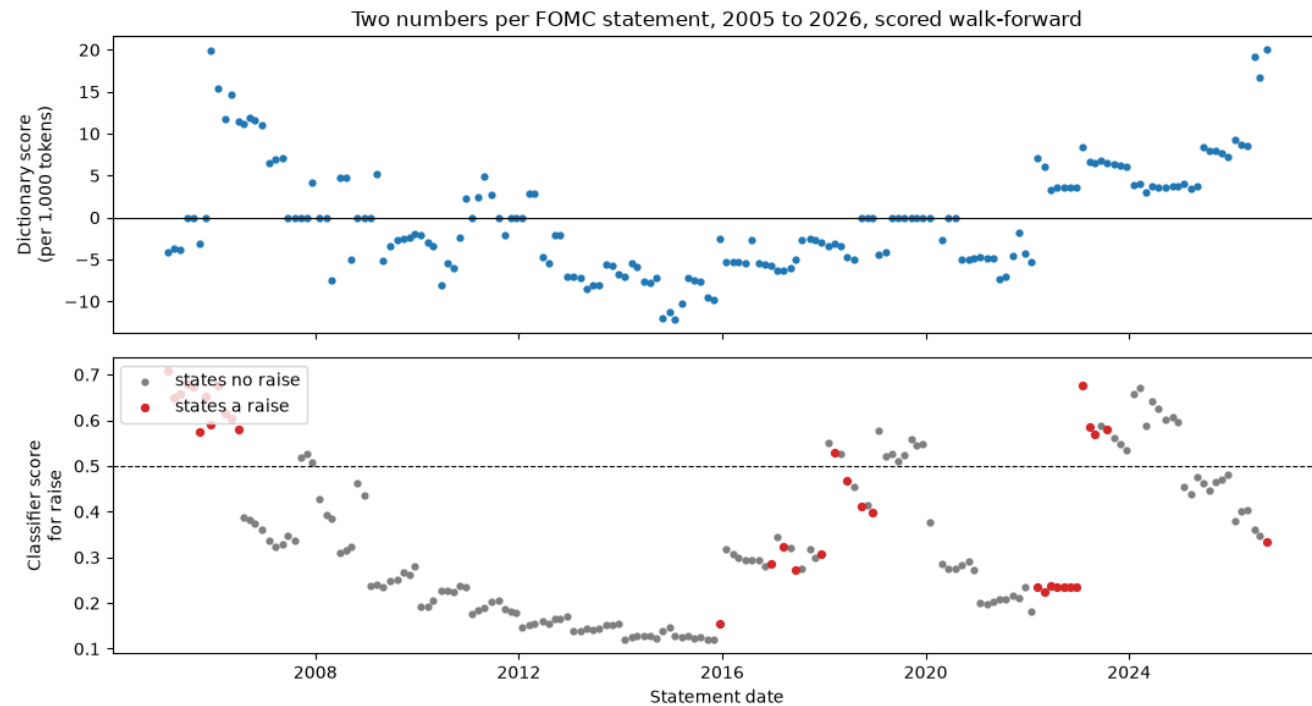
raise statements that hold it

18

other statements, mostly cuts

No pattern written for the first leak removes this one.

Side by side, by date



Same side of the medians: 0.6821 (both above 45, both at or below 73); rank correlation 0.5898. Scores move in year blocks.

What the model shows, and where it fails

0.6510

macro F1, interval 0.5615 to 0.7297

19 of 22

years labelled as one block

16 of 33

raises missed; 25 false flags

The label

policy_action read six 2015 statements as raise; the stated rate caught it

The leak

the related action on the discount rate is still in the text

The years

whole years one way: raise throughout 2005, 2019, 2023, 2024

The words

not lower still counts; raised and increases do not

The interval

statements of one year are alike: narrower than it should be

Both numbers describe the wording of statements already published. Source: FOMC post-meeting statements, Board of Governors of the Federal Reserve System, [federalreserve.gov](https://www.federalreserve.gov).

Excel side by side

Excel side by side (tested in Excel)

```
=SUMPRODUCT((LEN(LOWER(A2))-LEN(SUBSTITUTE(LOWER(A2),pos,"")))/LEN(pos))  
=SUMPRODUCT((LEN(LOWER(A2))-LEN(SUBSTITUTE(LOWER(A2),neg,"")))/LEN(neg))  
=(C2-D2)/B2*1000  
pos = H1:H5, neg = I1:I5 (named ranges)
```

2 on 2008-09-16; whole words 0
1; whole words 1
4.9505; whole words -4.9505
equal to str.count on all 213 rows

97

statements where the counts differ

268 / 194

firming: substrings / whole words

356 / 315

easing: substrings / whole words

"increasing" holds "easing"; "slower" holds "lower". No tokenizer in Excel.

Save the day: tag week 1

1. Commit

the modules and today's textkit

2. Log

results.csv, four decimals

3. Tag

git tag week1

4. Diff

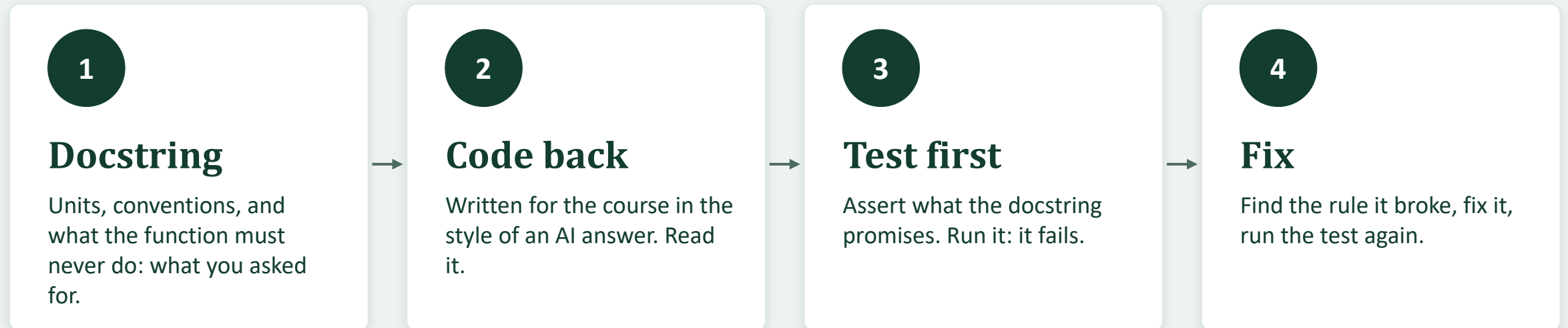
week1~1 against week1

```
week1
results.csv | 4 ++++
1 file changed, 4 insertions(+)
```

pytest: 30 passed. A tag names one commit for good.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.



Today: "Not" with a capital N is never a token: 0 of 213 scores change, though 26 texts hold "not".

Code that runs is not the same as code that is right.

What you can do now

- Write a word list, print it, and score per thousand tokens
- Test a label read from text against another reading
- Find the label in the text and take it out first
- Score dated documents walk-forward by year
- Count list words in Excel, and know what a substring is

Week 2, Day 1 Sentence embeddings and semantic search